

Proposed Solution Architecture: Neurosurgery Operative-Note Generation

Built from the starting stack — accurate segmentation model + endoscopy-domain VLM (Endo-FM/ZEN-style) + SurgGPT-style language head — adapted for neurosurgery per the feasibility analysis. This is a **modular, six-layer pipeline**, not a single end-to-end model, matching the "diagnose-before-training" and additive/feature-flagged discipline already established for this project.

Revision note. This version adds a **claim-level, object-grounded generation policy** trained with **GRPO + calibration rewards**. Segmented and tracked objects from Layer 3 get persistent IDs. Every factual claim in the note must cite those IDs, carry its own confidence score, and be verified against perception outputs during training and at inference. A dedicated section (Robustness to Unseen Data) covers how grounding holds up on new sites, instruments, anatomy and camera conditions.

Layer 1 — Input

Raw neurosurgery video (endoscopic pituitary/skull-base to start; microscope/exoscope for open cranial in a later phase).

Layer 2 — Adapted Perception Backbone

- **Vision backbone:** the existing endoscopy VLM (Endo-FM-class ViT-B/16 video transformer, or ZEN-style multi-teacher-distilled backbone), carried forward **not retrained from scratch**.
- **Adaptation applied:** DAPT (continued self-supervised pretraining on unlabeled neurosurgery video) → TAPT (task-specific unlabeled data) → PEFT (LoRA / Vector-MoLoRA) supervised fine-tuning. This sequence directly follows the staged plan from the earlier feasibility report and avoids catastrophic forgetting of the source-domain representation.
- **Camera/modality adapter:** a thin adapter layer handling the optics gap between rigid endoscope (Stage-1 target) and microscope/exoscope (Stage-2 target), including ICG-fluorescence mode.

Layer 3 — Structured Perception Modules (the "facts" the note is built from)

Three purpose-built heads sit on top of the adapted backbone — this is the biggest structural addition beyond fine-tuning:

1. **Phase/Step Recognizer** — new head, trained on a neurosurgery-specific ontology (PitVis-2023's step/instrument labels, or the Das et al. 27-step Delphi-consensus scheme for pituitary surgery). This supplies the temporal skeleton the note is organized around, since cholecystectomy-style 7-phase taxonomies don't transfer.

2. **Segmentation Model (adapted)** — your existing accurate segmenter, extended to neurosurgery classes via either (a) frozen-backbone + new head for known target classes, or (b) Grounded-SAM-2 (Grounding DINO + SAM2) for open-vocabulary prompting of novel/rare structures without exhaustive re-labeling.
3. **Instrument/Object Detector + Tracker** — new taxonomy covering neurosurgical instruments (dissectors, curettes, rongeurs, bipolar) and objects with safety relevance (cottonoids/patties), with identity tracking across occlusion (detect→track, à la ByteTrack/BoT-SORT or SAM2 mask tracking).

3.4 Object Registry (new) — the interface to the grounded generator

The three heads above are fused into a single **object registry**: one row per tracked entity with a **persistent ID**, which the language layer is required to cite.

Detect → track → re-detect loop. One detector (not two different models) runs on the first frame of each segment, then **re-runs every N frames and at every scope-exit/re-entry, lens-cleaning or camera cut**. New detections are matched to the masks SAM2 is currently propagating:

Match result	Action
High mask IoU with an existing track	Keep the ID; re-prompt SAM2 with the new box to correct drift
No overlap, appearance matches a recently lost track (ReID embedding)	Restore the old ID (e.g. instrument re-inserted after scope withdrawal)
No match	New ID

Registry row schema (per ID):

```
id | label (or "unknown") | source (closed-set head / open-vocab / class-agnostic)
  | time span(s) | per-frame masks | phase(s) present
  | perception_conf = f(detector score, closed/open-vocab label agreement, track stability, mask quality)
```

`perception_conf` is carried forward and used both as policy input and as an **evidence cap** on final claim confidence (Layer 5.3).

Unknown objects are first-class. Anything the class-agnostic proposer (SAM2 automatic masks / SAM 3 concept prompts) segments but no labeler can name confidently becomes `obj_k: unknown`. It is tracked and can be referred to, but the generator is trained to describe it as unidentified rather than invent a label (see Robustness section).

Benchmark choice for this layer: compare Grounding DINO + SAM2 against SAM 3 (single model that detects and tracks from text concept prompts) during the diagnosis stage, and keep whichever produces fewer ID switches on held-out pituitary video. ID consistency matters directly because the RL reward checks ID citations — tracker errors would otherwise be charged to the policy.

Layer 4 — Safety Branch (parallel, not sequential)

Runs alongside note generation rather than downstream of it, because it has a different optimization target (recall, not fluency):

- **RSI count-reconciliation module:** detect → track → maintain an in/out ledger for cottonoids/patties specifically (the dominant neurosurgical RSI class), tuned for high recall even at the cost of false positives.
- **Alert/flag:** surfaces directly to the surgeon, with a human sign-off gate before closure — this is a decision-support signal, not an autonomous stop/go decision.
- **Isolation from RL:** the ledger is deterministic and is **not** trained by the GRPO loop. The note generator may *cite* ledger entries (by ID) but cannot alter or override counts, so reward hacking in the narrative branch cannot leak into the safety signal.

Layer 5 — Language Generation Layer (grounded, claim-level, confidence-scored)

5.1 Inputs to the generator

- **Object tokens:** backbone features mask-pooled over each SAM2 mask, plus a learned ID embedding, so each registry entry becomes a token the model can attend to and cite. This is preferred over drawing ID numbers onto frames (Set-of-Mark overlays), because overlays occlude small anatomy in the endoscopic field and the backbone was never trained to read digits. Overlays remain an ablation.
- **Registry as text:** compact table (ID, label, time span, phase, perception_conf).
- **Phase skeleton** from the Phase Recognizer, and **RAG context** (terminology + exemplar notes).

5.2 Output format (structured, machine-checkable)

Per segment/phase, the generator emits reasoning, prose and a claim list:

```

<think>
obj_4 (Kerrison rongeur) is active 12:10-14:05 during sellar opening; obj_9
is
unlabeled, appears after obj_4 withdraws, mask adjacent to sellar floor ...
</think>
<caption>
The sellar floor was opened with a Kerrison rongeur [obj_4]. An unidentified
object [obj_9] was present near the opening from 14:10 ...
</caption>
<claims>
[{"text":"sellar floor opened with Kerrison rongeur","objs":[4],"t":
[730,845],
  "type":"tracker+phase","conf":0.88},
 {"text":"unidentified object near sellar opening","objs":[9],"t":[850,900],
  "type":"tracker","label":"unknown","conf":0.64}]
</claims>

```

(Illustrative content only.)

- `<think>` = the **object-grounded reasoning**: it may only reference registry IDs and phase labels, which makes each reasoning step auditable against Layer 3.
- `<type>` declares which perception module can verify the claim (see 5.4).
- **Anti-gaming rule**: an independent decomposer also splits the prose into claims and aligns them to the policy's list. Any prose claim missing from the list is scored with `conf = 1.0`, so the policy cannot hide uncertain statements from calibration.

5.3 Confidence scores

- **Claim confidence** (trained): `conf_i` emitted by the policy, calibrated by the RL rewards in 5.5.
- **Evidence cap** (inference-time, conservative): $\text{final_conf_i} = \min(T(\text{conf_i}), \max(\text{perception_conf over cited objects}))$, where T is temperature scaling fitted on a small labeled calibration set from the target site. A claim can never be more confident than the perception evidence behind it — this is the main safeguard on unseen data.
- **Section / caption confidence**: report both the mean and the minimum of `final_conf` over claims in the section; a section is flagged for surgeon review if **any claim marked critical** (complications, closure, retained-item references, anatomy injury) falls below threshold τ_{crit} , or the section mean falls below τ_{sec} .

5.4 Grounding/verification module (now also the reward source)

Replaces the original single cross-check with a **typed verifier**, used identically for RL rewards (training) and filtering (inference):

Claim type	Verified by	Deterministic?
Instrument/object presence & timing	Tracker registry (ID exists, mask present in claimed window)	Yes
Anatomy visible/exposed	Segmentation masks in claimed window	Yes
Step/phase membership	Phase Recognizer	Yes
Attributes/actions (e.g. "coagulated", "resected")	Surgical VQA verifier (PitVQA++-style LoRA model) on mask crops + frames in window	No — model-based
Counts / retained items	Safety-branch ledger only	Yes

Verifier outputs `(true / false / unverifiable)`; unverifiable claims are **masked out** of precision and calibration terms (never scored as correct). Deterministic checks are weighted above the model-based verifier, because a general or partially adapted VLM is the weakest link on surgical content.

5.5 Policy training: GRPO + claim-level calibration rewards

Policy: the SurgGPT-style head (LoRA) consuming 5.1 inputs; backbone and Layer 3 stay frozen during RL. For each segment, sample a group of G outputs (start G = 8, small KL coefficient) and compute:

Term	Definition	Purpose
<code>R_format</code>	0/1 gate; failure $\Rightarrow$ total reward 0	Parseable output
<code>R_ground</code>	Fraction of claims whose cited IDs exist and are present in the claimed window	Deterministic ID grounding
<code>P</code>	mean y_i over verifiable claims	Faithfulness
<code>Rec</code>	Coverage of reference claims + registry objects, weighted by clinical importance / prominence	Prevents "say less" collapse
<code>R_cal</code>	$-\text{mean}(\text{conf}_i - y_i)^2$ (Brier)	RLCR-style calibration; bounded, so GRPO advantages stay stable
<code>R_corrupt</code>	Penalty if confidence on claims about <code>obj_k</code> does not drop when <code>obj_k</code> 's evidence is degraded (RAC-style clean vs. corrupted pair)	Confidence tied to visual evidence
<code>R_unk</code>	Reward for labelling <code>unknown</code> registry objects as unidentified; penalty for asserting a specific label that the verifier rejects	Honest handling of novel objects

$$R = R_{\text{format}} \cdot [w_g \cdot R_{\text{ground}} + w_p \cdot P + w_r \cdot \text{Rec} + w_c \cdot R_{\text{cal}} + w_k \cdot R_{\text{corrupt}} + w_u \cdot R_{\text{unk}} - \lambda \cdot \text{length_penalty}]$$

Corruptions for `R_corrupt` are surgery-realistic, not ImageNet-style: mask-out/inpaint of `obj_k` via its SAM2 mask, synthetic smoke, blood occlusion, specular glare, lens fogging, motion blur. Applied on ~20% of batches (needs a second rollout).

Credit assignment: start with standard sequence-level GRPO; then move to **claim-level advantages** (per-claim reward normalized within the group, assigned to that claim's text + `conf` tokens; global terms spread over all tokens). Treat sequence- vs. claim-level as an explicit ablation.

Why RL is attached here and nowhere else: the phase, segmentation and tracking heads have supervised targets and stay supervised; the safety ledger stays deterministic. RL only shapes *what the generator says and how confident it is*, which is exactly where supervised training has no good target.

5.6 RAG module

Unchanged in role: retrieval over neurosurgical terminology and exemplar operative notes. Retrieved exemplars inform wording only; claims derived from RAG text rather than the

video are typed `unverifiable` and are excluded from the note's factual fields.

Layer 6 — Note Assembly

- **Hierarchical summarization:** clip-level → phase-level → full-procedure narrative, avoiding the long-context failure mode of asking a VLM to narrate a multi-hour video in one pass. Object IDs persist across levels via the registry's ReID, so a procedure-level claim can still cite the same instrument seen in several phases. Long-video RL uses a global score plus a randomly sampled segment score (VCap-style) so late-procedure content is not under-rewarded.
- **Structured template:** fields drawn from Royal College of Surgeons "Good Surgical Practice" (procedure, findings, complications, closure technique, etc.) — the same template pattern Das et al. used successfully for pituitary surgery, combining reliable structured fields with generative prose only where free text adds value.
- **Claim filtering:** only verified claims with `final_conf ≥ τ` enter structured fields automatically; claims below τ appear in the draft visibly marked as low-confidence rather than silently dropped.

Output — Human-in-the-Loop

Draft note with:

- per-claim confidence and the cited object IDs (clickable back to the video segment and mask),
- per-section confidence (mean and minimum) and review flags,
- the `<think>` grounding trace available on demand,
- safety-branch alerts merged in.

The surgeon reviews and signs off before the note becomes the clinical record. No autonomous publishing.

Robustness to Unseen Data (grounding that transfers)

Grounding must survive new hospitals, scopes, instruments, anatomy variants and unusual events. The design handles this at five points:

1. **Open-vocabulary + class-agnostic perception.** Closed-set heads cover known classes; open-vocabulary prompting (Grounded-SAM-2 or SAM 3) with vocabulary pulled from the RAG terminology corpus covers the rest; a class-agnostic mask proposer catches anything left, registered as `unknown`. Label agreement between closed-set and open-vocab sources feeds `perception_conf`.
2. **Perception uncertainty propagates.** Because final confidence is capped by `perception_conf`, a poorly detected or ambiguous object automatically yields low-confidence claims, even if the language head is overconfident on out-of-distribution input.

3. **Confidence trained to track evidence, not style.** R_{corrupt} with surgery-realistic degradations teaches the policy that confidence must fall when visual evidence falls — the property that matters most on unfamiliar video. R_{unk} teaches "unidentified object" as a legitimate answer.
4. **Target-site recalibration, cheaply.** Refit temperature scaling τ (and thresholds τ) on a small labeled set from each new site, without retraining the policy. Optionally use conformal risk control to set τ so that the rate of false claims in auto-filled fields stays below a chosen level on that site's calibration data.
5. **Label-free runtime checks.** Temporal consistency (claims about the same ID in adjacent segments must not contradict), registry consistency (every cited ID must exist), and verifier agreement are computed on every new video; failures lower section confidence and trigger review flags.

Evaluation protocol for "unknown data": leave-one-site-out and leave-one-scope-type-out splits; held-out instrument classes never seen by the closed-set head; synthetic corruption sweeps at increasing severity; for each, report claim precision/recall, grounding accuracy, per-claim ECE and AUROC, risk-coverage curves, and the confidence drop under object masking.

Why modular, not end-to-end

- Each layer can be validated and rolled back independently (matches the additive, feature-flagged discipline already used elsewhere in this project).
- The safety branch can be tuned for recall independently of the narrative branch's fluency/factuality tuning — conflating them risks either over-alerting (annoying) or under-alerting (dangerous).
- Swapping the camera/modality adapter is the only change needed to extend from endoscopic pituitary (Stage 1) to open cranial/microscope (Stage 2) — the rest of the pipeline is reusable.
- The RL loop touches only the generator; every reward term is computed from modules that are frozen and independently validated, so a failing reward component can be switched off by flag without destabilizing perception.

Staged rollout (ties to the earlier roadmap)

1. **Pituitary/eTSA only**, using PitVis/PitVQA/Das et al. resources — closest to the source endoscopy domain.
2. Validate Phase and Segmentation heads first (target: step-recognition $F1 \geq 0.80$, matching the Das et al. benchmark) before trusting the language layer's output.
3. **Diagnose the grounding stack before any RL:** measure tracker ID-switch rate; hand-audit ~200 typed-verifier judgments (fix the verifier if agreement is below ~85%); measure the base generator's zero-shot claim precision and the ECE of its verbalized confidence.

4. **SFT cold start** on a few thousand examples in the 5.2 output format (teach format and ID citation, not specific confidence values).
5. **GRPO, one reward at a time**, each behind a feature flag: (a) $\frac{R_{\text{format}} + R_{\text{ground}} + P + \text{Rec}}{4}$ → uncalibrated baseline; (b) add R_{cal} ; (c) add R_{corrupt} and R_{unk} ; (d) switch to claim-level advantages; (e) long-video global + random-segment scoring. Track reward, response length and recall from the first step to catch collapse toward short, safe notes.
6. Add RAG + verification at inference, then pilot the RSI branch separately (it has its own data/annotation needs — cottonoid detection isn't covered by any existing instrument dataset).
7. Run the unseen-data evaluation protocol above before any clinical pilot.
8. Only after Stage 1 works end-to-end, extend to open cranial via the camera/modality adapter, recalibrating τ and τ on the new modality.

Key ablations

No object IDs (text-only claims) · ID overlays vs. mask-pooled object tokens · without R_{cal} · without R_{corrupt} · without R_{unk} · Brier vs. log-score calibration reward · sequence- vs. claim-level advantages · with vs. without the evidence cap · Grounding DINO + SAM2 vs. SAM 3 in the registry.